

WASP-135b

R-1332

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_210602 · created 2026-07-31T20:41:45+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459367.8501 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.032
Transit depth (Rp/Rs)^2	3.05 +/- 1.11 [%]
Orbital Inclination (inc)	81.68 +/- 1.78
Airmass coefficient 1 (a1)	1.293 +/- 0.035
Airmass coefficient 2 (a2)	-0.24 +/- 0.15
Scatter in the residuals of the lightcurve fit is	2.73 %
Best Comparison Star	#3 - [367, 197]
Optimal Aperture	3.5
Optimal Annulus	9.94
Transit Duration (day)	0.067 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.175 $\pm$ 0.032 vs 0.139 (deviation 1.1 σ)
Duration fit vs published	0.067 d vs 0.0687 d (deviation 0.11 σ)
Mid-time O-C	-14.1 min
Depth SNR	5.4
Residual scatter	2.73 %

Light curve

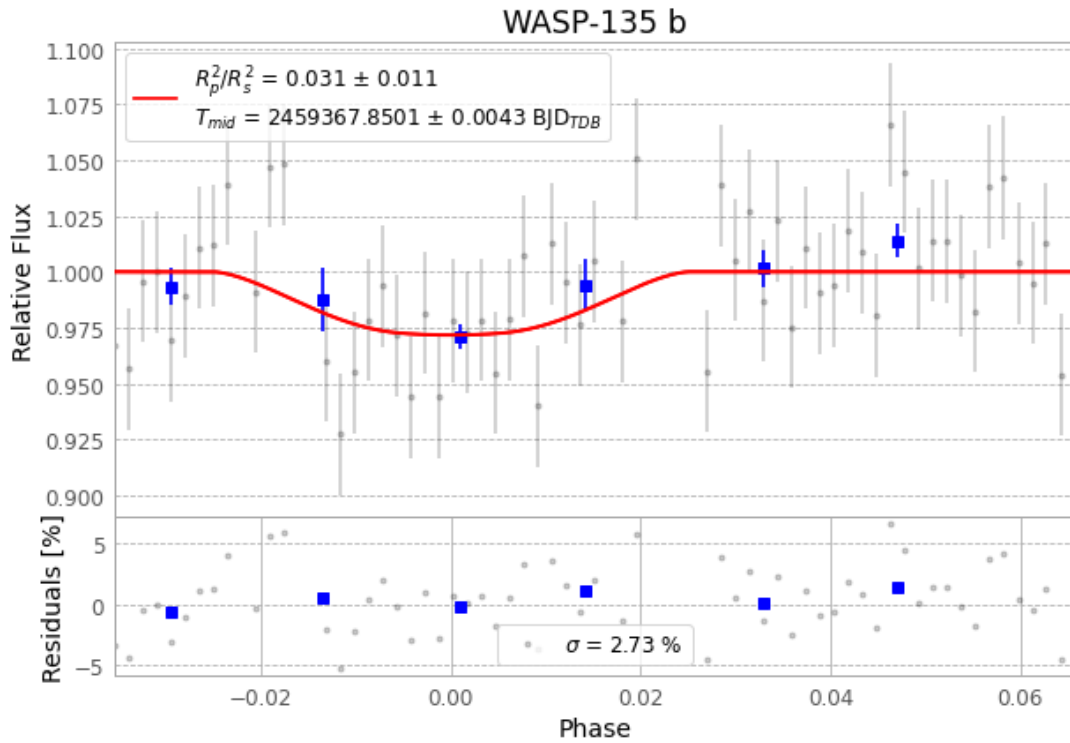


Figure 1 — FinalLightCurve_WASP-135 b_2021-06-02.png (SHA-256 9a04ad0bef3c97ab...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [363, 180], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 89bf112f7e3bbf6d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), WASP-135210602070612.FITS → WASP-135210602103012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-210602.log (2f47bef4f225...), FinalLightCurve_WASP-135 b_2021-06-02.png (9a04ad0bef3c...), FinalLightCurve_WASP-135 b_2021-06-02.csv (c244d41bcf61...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 20:41Z.